

Risk Neutral Valuation: Two-Horse Race Example

- One horse has 20% chance to win another has 80% chance
 - \$10000 is put on the first one and \$50000 on the second
- If odds are set 4-1:
- Bookie may gain \$10000 (if first horse wins)
 - Bookie may lose \$2500 (if second horse wins)
 - Bookie expects to make $0.2 * (10000) + 0.8 * (-2500) = 0$
- If odds are set 5-1:
- Bookie will not lose or gain money no matter which horse wins

Morgan Stanley

Forward Contract $\rightarrow$

Will buy asset S at time T in future for price P .

If $P \uparrow$ profit for buyer
 $P \downarrow$ " " seller

$\Rightarrow$ Call option,

You buy the right to buy asset at certain time.

Premium

$\Rightarrow$ Put option

Same but to sell

- ⇒ Given current price of the stock and the assumptions on the derivative of stock price, there is no uncertainty about the price of derivative.
- ⇒ The Price is defined only by the price of the stock and not by the preferences of market participants.
- ⇒ Mathematical apparatus allows to compute current price of a derivative and its risks, given certain assumptions about the market.

Risk Neutral Valuation: One step binomial tree

$S \rightarrow$ stock riskless money market account $\rightarrow B$
 $r \rightarrow$ rate of int. derivative claim f



- ⇒ Forward contract, pays $S - K$ in time dt .
 $K \rightarrow$ should be defined by real world.

$$p(s_1 - K) + (1-p)(s_2 - K) = p s_1 + (1-p)s_2 - K$$

$\uparrow$
Individual
 $\uparrow$
collective

i f $k = 1/2$, $K = \frac{S_1 + S_2}{2}$

Ex.

1. Borrow \$ S_0 to buy a stock. Enter Forward contract with strike K_0 .
2. In time dt deliver stock in exchange for K_0 and repay \$ $S_0 e^{rdt}$

If $S_0 e^{r_d t} < K_0$ riskless profit
 $> K_0$ Always loss

$$S_0 \quad S_0 e^{rdt} = K_0 \quad * * *$$

$$\Rightarrow \begin{aligned} f_1 &= a S_1 + b B_0 e^{\gamma_1 \Delta t} \\ f_2 &= a S_2 + b B_0 e^{\gamma_2 \Delta t} \end{aligned}$$

Then, $f_0 = a S_0 + b B_0$

$$a = \frac{f_1 - f_2}{s_1 - s_2} \quad b = \frac{s_1 f_2 - s_2 f_1}{(s_1 - s_2) B_0 e^{\gamma dx}}$$

